

QUESTION 9

TITLE:

Frequency Response Analysis of $H(s) = 2s/(s^2 + 2s + 25)$

AIM:

To define the transfer function $H(s) = 2s/(s^2 + 2s + 25)$ in MATLAB, calculate its frequency response over 0.1 to 100 rad/s, plot the magnitude and phase responses, and determine the centre frequency and -3 dB bandwidth.

SOFTWARE REQUIRED:

MATLAB with Control System Toolbox.

THEORY:

The given transfer function is

$$H(s) = 2s/(s^2 + 2s + 25)$$

For frequency response, put $s = j\omega$.

$$H(j\omega) = 2j\omega/(25 - \omega^2 + j2\omega)$$

The magnitude response is

$$|H(j\omega)| = 2\omega / \sqrt{(25 - \omega^2)^2 + 4\omega^2}$$

The centre frequency is approximately

$$\omega_c = 5 \text{ rad/s}$$

The -3 dB frequencies are approximately

$$\omega_1 = 4.10 \text{ rad/s}$$

$$\omega_2 = 6.10 \text{ rad/s}$$

Therefore,

$$\text{BW} = \omega_2 - \omega_1$$

$$\text{BW} \approx 6.10 - 4.10$$

$$\text{BW} \approx 2.00 \text{ rad/s}$$

PROCEDURE:

1. Define the Laplace variable s.
2. Define the transfer function using tf().
3. Generate the frequency range from 0.1 to 100 rad/s.
4. Obtain the frequency response using freqresp().
5. Calculate magnitude and phase.
6. Convert magnitude into dB.
7. Find the peak magnitude and centre frequency.
8. Find the lower and upper -3 dB frequencies using find().
9. Calculate the bandwidth.
10. Plot the magnitude and phase responses.
11. Display the calculated results.

MATLAB PROGRAM:

```
clc;

clear;

close all;

s = tf('s');

% Define transfer function

H = (2*s)/(s^2 + 2*s + 25);

% Frequency range

w = logspace(-1, 2, 2000);

% Frequency response

[Hjw, wout] = freqresp(H, w);

Hjw = squeeze(Hjw);

% Magnitude and phase

mag = abs(Hjw);

mag_dB = 20*log10(mag);

phase = angle(Hjw)*180/pi;

% Find peak magnitude and centre frequency

[max_mag, peak_index] = max(mag);

centre_frequency = wout(peak_index);

peak_dB = 20*log10(max_mag);
```

% -3 dB level

level_3dB = peak_dB - 3;

% Lower -3 dB frequency

lower_index = find(mag_dB(1:peak_index) <= level_3dB, 1, 'last');

w1 = wout(lower_index);

% Upper -3 dB frequency

upper_temp = find(mag_dB(peak_index:end) <= level_3dB, 1, 'first');

upper_index = peak_index + upper_temp - 1;

w2 = wout(upper_index);

% Bandwidth

BW = w2 - w1;

% Display results

fprintf('\nFREQUENCY RESPONSE RESULTS\n');

fprintf('-----\n');

fprintf('Peak Magnitude = %.4f\n', max_mag);

fprintf('Peak Magnitude = %.4f dB\n', peak_dB);

fprintf('Centre Frequency = %.4f rad/s\n', centre_frequency);

fprintf('Lower -3 dB Frequency = %.4f rad/s\n', w1);

fprintf('Upper -3 dB Frequency = %.4f rad/s\n', w2);

fprintf('-3 dB Bandwidth = %.4f rad/s\n', BW);

```
% Plot magnitude and phase

figure;

subplot(2,1,1);

semilogx(wout, mag_dB, 'LineWidth', 1.5);

grid on;

hold on;

plot(centre_frequency, peak_dB, 'ro');

plot(w1, level_3dB, 'ko');

plot(w2, level_3dB, 'ko');

xlabel('Frequency (rad/s)');

ylabel('Magnitude (dB)');

title('Magnitude Response');

subplot(2,1,2);

semilogx(wout, phase, 'LineWidth', 1.5);

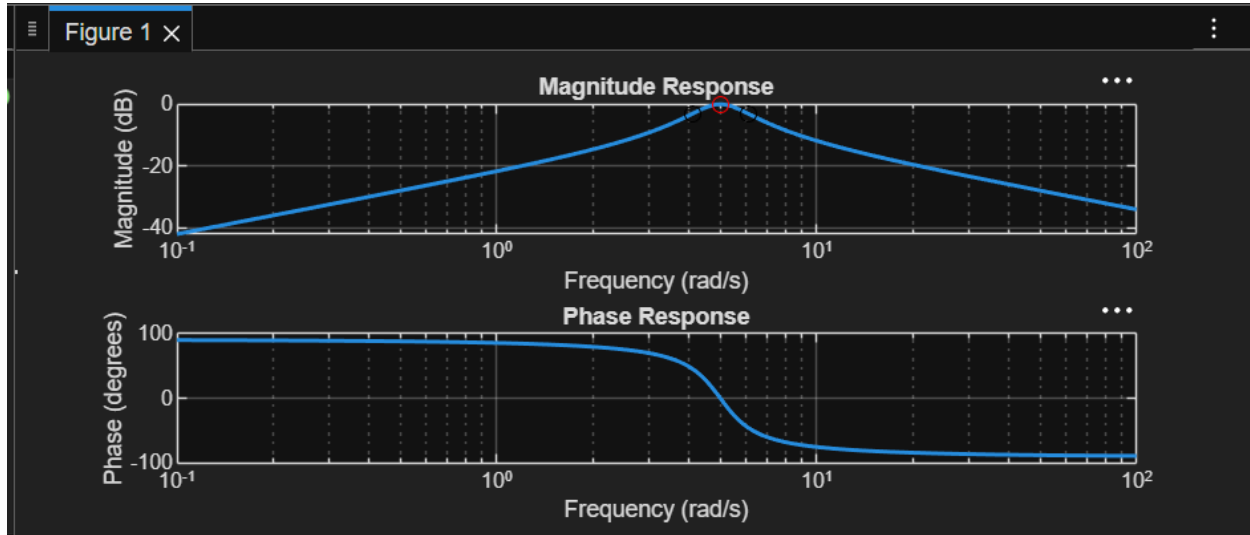
grid on;

xlabel('Frequency (rad/s)');

ylabel('Phase (degrees)');

title('Phase Response');
```

OUTPUT:



ANALYSIS:

The magnitude response reaches its maximum value near 5 rad/s. The peak magnitude is approximately 1 or 0 dB.

The -3 dB frequencies are approximately 4.1 rad/s and 6.1 rad/s. Hence, the bandwidth is approximately 2 rad/s.

The system attenuates frequencies below and above the centre frequency and has maximum response around 5 rad/s.

RESULT:

Centre frequency = 5 rad/s

-3 dB bandwidth = 2 rad/s

CONCLUSION:

The frequency response of the given transfer function was successfully analyzed using MATLAB. The centre frequency and -3 dB bandwidth were obtained from the simulated frequency response.

QUESTION 29

TITLE:

Nyquist Stability Analysis and Gain Margin Design

AIM:

To plot the Nyquist diagram of $G(s) = 5/[(s+2)(s+5)(s+1)]$ using MATLAB, determine the gain margin and phase margin using `margin()`, design a gain K to obtain a gain margin of 6 dB, and verify the result using the Nyquist plot.

SOFTWARE REQUIRED:

MATLAB with Control System Toolbox.

THEORY:

The given open-loop transfer function is

$$G(s) = 5/[(s+2)(s+5)(s+1)]$$

The poles are

$$s = -1, -2, -5$$

All poles are in the left-half s-plane. Therefore, the open-loop system has no right-half-plane poles.

According to the Nyquist stability criterion, closed-loop stability is determined by observing the encirclement of the critical point

$(-1, 0)$

For a gain margin of 6 dB,

$$\text{GM} = 10^{(6/20)}$$

$$\text{GM} \approx 1.995$$

The phase crossover frequency is approximately

$$\omega_{pc} = \sqrt{17}$$

$$\omega_{pc} \approx 4.123 \text{ rad/s}$$

At this frequency,

$$|G(j\omega_{pc})| \approx 0.03968$$

Therefore,

$$K = 1/[GM \times |G(j\omega_{pc})|]$$

$$K \approx 12.6$$

PROCEDURE:

1. Define the open-loop transfer function using tf().

- 2. Plot the Nyquist diagram using nyquist().**
- 3. Mark the critical point (-1,0).**
- 4. Calculate gain margin and phase margin using margin().**
- 5. Determine the phase crossover frequency.**
- 6. Convert the required 6 dB gain margin into absolute gain margin.**
- 7. Calculate the required gain K.**
- 8. Form the modified open-loop transfer function KG(s).**
- 9. Plot the modified Nyquist diagram.**
- 10. Calculate the new gain margin and phase margin.**
- 11. Analyze the stability of the modified system.**

MATLAB PROGRAM:

```
clc;
```

```
clear;
```

```
close all;
```

```
s = tf('s');
```

```
% Open-loop transfer function
```

```
G = 5/((s+2)*(s+5)*(s+1));
```

% Original Nyquist plot

figure;

nyquist(G);

grid on;

hold on;

plot(-1,0,'rx','MarkerSize',10,'LineWidth',2);

xlabel('Real Axis');

ylabel('Imaginary Axis');

title('Nyquist Plot of Original System');

% Calculate original margins

[Gm,Pm,Wcg,Wcp] = margin(G);

fprintf('\nORIGINAL SYSTEM\n');

fprintf('-----\n');

fprintf('Gain Margin = %.4f dB\n',20*log10(Gm));

fprintf('Phase Margin = %.4f degrees\n',Pm);

fprintf('Phase Crossover Frequency = %.4f rad/s\n',Wcg);

```
fprintf('Gain Crossover Frequency = %.4f rad/s\n',Wcp);
```

```
% Required gain margin
```

```
GM_required_dB = 6;
```

```
GM_required = 10^(GM_required_dB/20);
```

```
% Phase crossover frequency
```

```
wpc = sqrt(17);
```

```
% Frequency response at phase crossover
```

```
Gjw = squeeze(freqresp(G,wpc));
```

```
% Magnitude at phase crossover
```

```
mag_pc = abs(Gjw);
```

```
% Calculate required gain
```

```
K = 1/(GM_required*mag_pc);
```

```
fprintf('\nGAIN DESIGN\n');
```

```
fprintf('-----\n');

fprintf('Required Gain K = %.4f\n',K);


% Modified open-loop transfer function

GK = K*G;


% Modified Nyquist plot

figure;

nyquist(GK);

grid on;

hold on;

plot(-1,0,'rx','MarkerSize',10,'LineWidth',2);

xlabel('Real Axis');

ylabel('Imaginary Axis');

title('Nyquist Plot After Gain Adjustment');


% Verify new margins

[Gm_new,Pm_new,Wcg_new,Wcp_new] = margin(GK);
```

```

fprintf('\nMODIFIED SYSTEM\n');

fprintf('-----\n');

fprintf('Gain K = %.4f\n',K);

fprintf('Gain Margin = %.4f dB\n',20*log10(Gm_new));

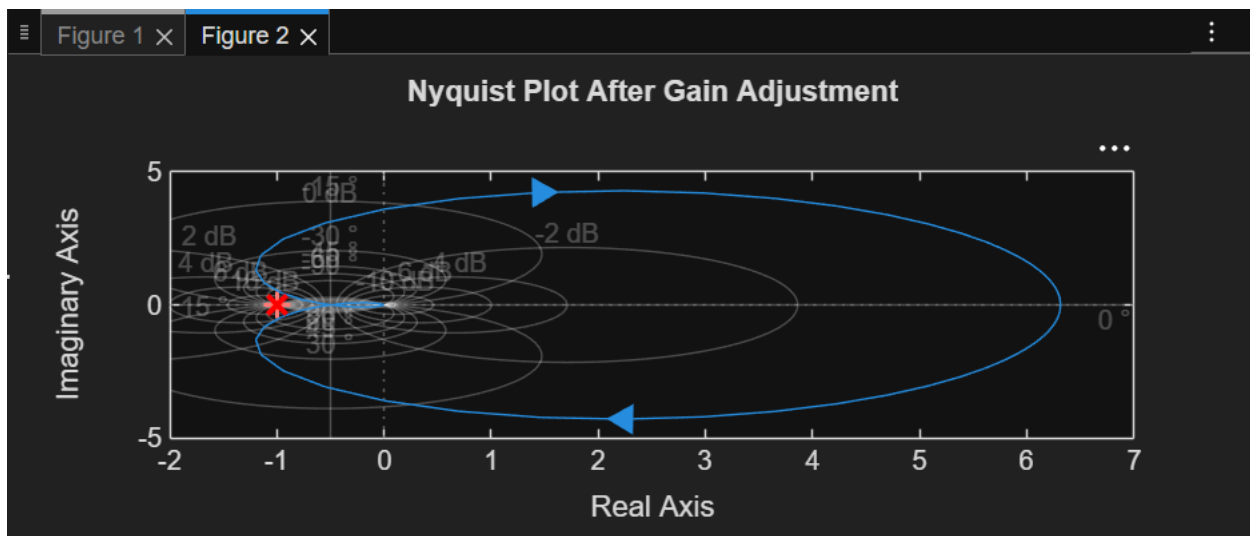
fprintf('Phase Margin = %.4f degrees\n',Pm_new);

fprintf('Phase Crossover Frequency = %.4f rad/s\n',Wcg_new);

fprintf('Gain Crossover Frequency = %.4f rad/s\n',Wcp_new);

```

OUTPUT:



ANALYSIS:

The original system has three poles at -1, -2 and -5. Therefore, there are no right-half-plane open-loop poles.

The original Nyquist plot does not encircle the critical point $(-1,0)$, indicating that the unity-feedback closed-loop system is stable.

The original gain margin is approximately 28 dB. To obtain the required gain margin of 6 dB, the gain is increased to approximately 12.6.

Increasing the gain expands the Nyquist plot towards the critical point. The modified system has a gain margin of approximately 6 dB.

The phase crossover frequency remains unchanged because a constant gain changes the magnitude but does not change the phase.

RESULT:

Required gain:

$$\mathbf{K \approx 12.6}$$

Gain margin after gain adjustment:

$$\mathbf{GM \approx 6 \text{ dB}}$$

The closed-loop system remains stable.

CONCLUSION:

The Nyquist stability analysis of the given system was successfully performed using MATLAB. The gain and phase margins were calculated using `margin()`. A gain of approximately 12.6 was designed to obtain the required 6 dB gain margin, and the modified Nyquist plot confirmed system stability.